

Can WANG

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Education

The Hong Kong Polytechnic University	MSc in Urban Informatics and Smart Cities	2022.9 - 2024.10
Capital Normal University	BSc in Geographic Information Science	2018.9 - 2022.6

AWARDS: OUTSTANDING GRADUATE (2022), UNIVERSITY FIRST-CLASS SCHOLARSHIP (2021), UNIVERSITY THIRD-CLASS SCHOLARSHIP (2020)

Research Interests

Multimodal Learning, Visual Grounding and Reasoning, AI-Generated Image/Video Forensics, Image/Video Generation

Academic Papers

LaP-Forensics: Latent-Pixel Consistency Guided Multimodal Reasoning for Deepfake Detection [arXiv:2607.25962]	FIRST AUTHOR, CCF-A (ACM MM 2026)
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#Deepfake Detection/Localization, #MLLM, #DDIM, #SAM, #SFT, #GRPO 2026

- **Problem & Idea:** Addressed reasoning hallucination in MLLM-based AI-generated image forensics, where visually realistic synthetic images can trigger plausible but false artifact explanations; proposed grounding forensic reasoning in latent-pixel consistency that beyond surface RGB cues.
- **Method:** Designed a dual-stream framework that fuses RGB semantic features with DDIM inversion-reconstruction residuals; structured Where-What-Why CoT reasoning produces [SEG] representations that are converted into prompts for SAM-based artifact localization.
- **Result:** Achieved 72.19% mIoU / 63.62% F1 on SynthScars, with +9.14 mIoU over SFT-only and +10.36 mIoU over the w/o forensic-stream variant; the standalone detector improved average UniversalFakeDetect accuracy by +8.24 points over baseline method.

Project/Research Experience

AI Dance Video Pose Transfer Generation Based on Wan2.1	RESEARCH PROJECT, INIT:AI CO.
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#Wan2.1, #LoRA, #DiffSynth, #Diffusion Video 2025.8 - 2025.10

- Built a 5,000+ video dance dataset and fine-tuned Wan2.1-VACE with DiffSynth/LoRA. Achieved an improved pose fidelity and identity consistency with the fine-tuned model

Real-Time 3D Hand Pose Estimation via Asynchronous RGB-Event Fusion	RESEARCH PROJECT, DVSENSE CO.
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#3D Reconstruction, #Pose Estimation, #Event Camera, #Multimodal Fusion, #MANO 2024.8 - 2025.1

- Designed an asynchronous RGB-event fusion framework with a cached slow branch for multi-scale frame-guided hand structure features and a fast event-buffer branch for high-frequency motion cues between RGB frames.
- Combined deformable spatiotemporal alignment with MANO-based two-hand parameter prediction to produce temporally responsive 3D joints and meshes, improving MPJPE by 5 mm / 52% over a single-RGB baseline while reducing inference latency by 20 ms.

Building Change Detection on Panoramic Street View Images	MASTER'S GRADUATION DISSERTATION
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#SVCD, #YOLO, #ViT, #Feature Matching, #End-to-End Detection 2023.8 - 2024.4

- Formulated street-view building change detection as instance-level visual inconsistency localization under viewpoint shift and occlusion; built a 3k+ image-pair / 10k+ instance benchmark with Siamese ViT-feature matching, improving recall by 37% and accuracy by 16%.

Appointments

AIGC Algorithm Development (Remote)	INIT:AI CO.	2025.8 - 2025.10
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Algorithm Engineer	DVSENSE (BEIJING) TECHNOLOGY CO., LTD.	2024.8 - Present
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- Conducted multimodal RGB-event reconstruction research, including dataset construction, model validation, and robustness evaluation for challenging real-world hand-motion scenes.
- Development of AI Customer Service based on Qwen3 with RAG, including retrieval system design, knowledge base construction, and response quality evaluation.

Research Assistant	OTTO POON FUND INSTITUTE FOR SMART CITIES, THE HONG KONG POLYTECHNIC UNIVERSITY	2022.8 - 2023.8
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- GBA Big Data Platform: Land-Use Category Conversion System via large-scale RS Image Segmentation(#DL, #Qt); Development of Data Annotation System

Skills

Models & Methods	MLLMs, Diffusion Models, Transformers, CNNs, CLIP, SAM, LoRA, SFT, GRPO, RAG
ML Platforms	PyTorch, HuggingFace Transformers
Programming & Systems	Python, C++, LaTeX, Linux, Git, Docker, ComfyUI
Languages	Mandarin, English (IELTS 7; CET4/6: 550+), Basic Cantonese and French
Multimedia	Photography, Video Editing, Graphic Design, AI-Aided Automation Pipeline for Media Production
Hobbies	Basketball, Snowboarding, Cycling, Kayaking